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Charity Ratings

Barış K. Yörük
Department of Economics
University at Albany
State University of New York
Albany NY 12222
byoruk@albany.edu

How do charitable donors respond to the third-party ratings that signal the quality of charities? I investigate this question using a novel data set from Charity Navigator, which provides quality ratings for 5,400 charities. Because Charity Navigator prominently displays a charity's star rating which is assigned based on its overall rating, one can identify the causal impact of a one star increase in ratings on charitable contributions with a regression discontinuity framework that exploits the threshold values of the overall ratings. I find that in general, the third-party ratings have a minor and often insignificant impact on charitable contributions received by charities. However, for relatively small charities, a higher rating leads to an increase in charitable contributions received. In particular, for these charities, I find that a one star increase in ratings is associated with a 19.5% increase in the amount of charitable contributions received. This result is robust under alternative model specifications and highlights the role of the third-party ratings in not-for-profit markets.

Key words: *charitable contributions; charity ratings*

1. Introduction

Many markets are characterized by incomplete information. When consumers cannot observe the true qualities of products or services that they intend to purchase, they cannot make informed decisions. In recent years, this classic market failure led to the proliferation of online consumer review and product rating web sites that provide information about the quality of products and services. The effect of the third-party ratings or consumer reviews on markets and consumers' decisions is ambiguous. On the one hand, they may complement or substitute for existing information sources such as advertising and help consumers to make informed decisions. On the other hand, reviews or ratings may be subjective or hard to interpret and cover only a small segment of a market. The existing literature mostly focuses on the role of consumer reviews and independent ratings in for-profit sectors. For instance, Luca (2011) investigates the effect of online consumer reviews on restaurant demand. Using data from Yelp.com, he finds that a one star increase in online rating leads to a 5–9% increase in restaurant revenues. Jin and Sorensen (2006) show that publicized health plan ratings have a significant impact on individuals' health plan choices, particularly for individuals choosing a plan for the first time. Varkevisser et al. (2012) find that patients have a high propensity to choose hospitals with a good reputation measured by publicly available quality ratings.

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Similarly, recent studies also show that expert reviews may play an important role in consumers' decision making process. For example, Ali et al. (2008) document the positive impact of expert ratings on wine prices and Reinstein and Snyder (2005) show that positive expert reviews have a significant effect on the box office revenue of movies.

The impact of expert reviews or the third-party ratings on the services and products provided by nonprofit organizations are relatively unknown.[1] Few studies investigate the effect of the third-party ratings on charitable giving. Using panel data of charities from American Institute of Philanthropy, Chhaochharia and Ghosh (2008) find that compared to charities with the lowest ratings, charities with the highest rating receive 16% more charitable donations. Using data from 405 charities rated by Charity Navigator, Gordon et al. (2009) find a 39% increase in private donations for a one star increase in Charity Navigator ratings. Sloan (2009) uses nonprofit ratings for New York charities from the Better Business Bureau's (BBB) Wise Giving Alliance to estimate the impact of accountability ratings on the amount of contributions an organization receives. She finds that the Wise Giving Alliance "pass" ratings have a statistically significant effect on the contributions received; however, the effect of "did not pass" ratings are insignificant. Using data from New York charities, Chen (2009) investigates the effect of BBB standards on giving. She finds that the BBB standards have a positive effect on giving behavior and meeting the BBB standards is associated with higher levels of public support. Using a sample of charities rated by Charity Navigator, Grant (2010) finds that a one star increase in ratings is associated with a 3–8% increase in contributions received by charities. She also shows that upon being rated, charities tend to receive fewer contributions than before. On the other hand, using data on 51 charities that operate in the State of Washington, Szper and Prakash (2011) find no significant relationship between charity ratings and donor contributions. Similarly, Frumkin and Kim (2001) find that nonprofits that position themselves as cost efficient by reporting low administrative to total expense ratios fared no better over time than less efficient appearing organizations in the market for individual, foundation, and corporate contributions. They argue that economizing may not always be the best strategy in the nonprofit sector.

This paper contributes to the existing literature which investigates the effect of the third-party ratings on charitable behavior. I use data from Charity Navigator, the largest independent charity evaluator in the United States. Compared with Gordon et al., my sample is relatively larger and includes 5,400 charities that are assigned an annual quality rating during 2007–2010. In order to identify the effect of the third-party ratings on charitable giving, previous studies mostly relied on either cross-sectional data and simple ordinary least squares (OLS) estimation or panel data models using fixed effects and first differences estimators. Simple OLS estimation fails to control for potential endogeneity of third-party ratings in charitable giving models. Similarly, time-variant unobservable charity characteristics that are correlated with charitable contributions may also be correlated with the third-party ratings. In this case, panel data models may provide a biased estimate of the effect of third-party ratings on charitable behavior.

Because Charity Navigator prominently displays a charity's star rating, which is assigned based on its actual overall rating, one can identify the causal impact of a one star increase in ratings on charitable contributions with a regression discontinuity (RD) framework that exploits the threshold values of the overall ratings. The qualities

1. An exception is a recent paper by Luca and Smith (2013). They show that when explicit rankings of colleges are published in U.S. News, a one-rank improvement leads to a 1-percentage-point increase in the number of applications to that college.

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of charities that are slightly below or above the threshold values of overall ratings should be very similar. [2] Therefore, differences in the amount of charitable contributions received by charities that are slightly below or above the threshold values of the overall ratings should be solely due to the one star difference in the corresponding star ratings. Using the RD design, I find that a one star increase in Charity Navigator ratings has no significant effect on the amount of charitable contributions received by charities. This result contradicts with the findings of previous literature. In order to test whether the effect of third-party ratings differ by charity size, I also estimate separate models for relatively small or large charities. Although the effect of ratings on large charities is statistically insignificant, the third-party ratings have a considerable impact on the amount of charitable donations received by relatively small charities. For this group, a one star increase in the rating is associated with a 19.5% increase in the amount of charitable contributions received. However, I do not have access to detailed data on Charity Navigator's web site traffic and cannot able to determine whether this effect is due to more web site traffic to Charity Navigator for small charities or a greater impact of Charity Navigator's small charity ratings for a fixed number of web site visitors. I also find some evidence that the effect of an increase in the star rating is significantly higher for charities that have relatively higher ratings. These results are robust to alternative model specifications.

The rest of the paper is organized as follows: The next section introduces the data. Section 3 provides a brief discussion of the empirical methodology. Section 4 presents the main results and discusses the sensitivity of these findings under alternative model specifications. Section 5 provides a discussion of policy implications and concludes.

2. Data

Charity Navigator is the largest independent charity evaluator in the United States and publishes its assessment of selected charities online at www.charitynavigator.org. The online traffic to this web site is the main indicator of the potential popularity of Charity Navigator's ratings and an important factor in determining whether or not these ratings have a considerable impact on charitable contributions. It seems that Charity Navigator is known by millions. Grant (2010) shows that from January 2004 to October 2009, Charity Navigator's web site averaged 50,000–120,000 visits per week. Charity Navigator's self-assessment of annual web site traffic in 2009 is more than 3 million unique visitors. The web site has also preserved its popularity over time. According the most recent estimate, in February 2014, 214,000 unique visitors viewed Charity Navigator's web site. The web site also gets a lot of attention during Decembers, which is the end of the tax year and coincides with the winter holidays. For instance, in December 2013, more than 450,000 people visited the web site.[3] Charity Navigator also receives media attention and is regularly mentioned on network TV and in articles published in popular outlets such as *The New York Times* and *Wall Street Journal*.

2. Overall rating is a measure of the quality for charities. Overall ratings are calculated based on the observable characteristics of charities such as fundraising expenses, administrative costs, program service revenues, and charitable contributions received.

3. There are several web sites that provide an analysis of web site traffic. For instance, for an analysis of web site traffic to Charity Navigator's web site, one can refer to www.compete.com. For comparison purposes, in May 2014, 40,225 unique visitors viewed Charity Watch, a web site that also provides third-party ratings of charitable organizations. Other third-party rating web sites attract much more people. During the same period (May 2014), Yelp.com was visited by more than 27 million people. BBB was visited by more than 4 million people and Consumer Reports was visited by more than 3 million people.

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Table I.
Summary of Charity Ratings

	Overall Rating				
	$R < 25$	$25 \leq R < 40$	$40 \leq R < 50$	$50 \leq R < 60$	$60 \leq R$
Star rating	0	1	2	3	4
Qualitative rating	Exceptionally poor	Poor	Needs improvement	Good	Exceptional
Number of obs.					
Full sample	170	1,259	3,383	6,136	4,685
Small charities	48	516	1,269	1,763	1,298
Large charities	10	150	776	2,224	1,794

Notes: R represents the actual overall rating, a number between 0 and 70, where a higher number represents better quality.

The information on the demographics of visitors of the Charity Navigator's web site is relatively limited. According to the estimates from Alexa, an online platform that provides demographic information on web site visitors, females and well-educated people are more likely to visit Charity Navigator's web site. This is consistent with the findings from the existing literature, which imply that compared with males, females are not only more likely to donate but also relative to their incomes, they contribute more to charity (Piper and Schnepf, 2008; Yörük, 2010).

Charity Navigator evaluates organizations granted tax-exempt status under section 501(c) (3) of the Internal Revenue Code and that file a Form 990, which excludes private foundations and many religious organizations. In order to be rated by Charity Navigator, a charity should satisfy several conditions. First, it should have received more than $500,000 in charitable donations and had a total revenue more $1,000,000 in the most recent fiscal year. Second, the main component of its revenues should be donations from individual givers. Charities that receive most of their funding from government grants or from fees they charge for their programs and services and that do not spend any money for fundraising are not rated.[4] Charity Navigator ratings are based on financial information from annual tax returns of charities. The complete formula used in the calculation of ratings is explained in detail at Charity Navigator's web page. Basically, this calculation includes metrics of ratios of operating and administrative costs, fundraising and program expenses, and the amount of donations received. These metrics are summarized and each charity is assigned an overall or actual rating, a number between 0 and 70. Based on the threshold values of the overall rating, each charity is given a star and a corresponding qualitative rating, where each additional star represents an increase in the quality level as summarized in Table I. Ratings are reviewed annually and a charity may receive a new rating based on the information in its tax return in the most recent fiscal year. The ratings calculation remains the same through all years and in general, higher fundraising and administration ratios hurt ratings, whereas increases in the other determinants of ratings improve the rating score.

Compared with the overall ratings, star and the associated qualitative ratings are relatively easy to interpret. Furthermore, Charity Navigator appears to be heavily

4. One can argue that among hundreds of thousands of charities that are required to file a tax return, those that are rated by the Charity Navigator are relatively large in terms of their revenues and public support. However, charities that are rated by the Charity Navigator account for roughly 50% of the donations in the United States. I acknowledge that the findings of this paper cannot be generalized to all charities due to the restrictions of the data set. However, the results provide useful insights to charitable donors and nonprofits about the value of third-party ratings.

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advertising the star ratings rather than the actual overall ratings. For instance, the main page of the Charity Navigator's web site has a direct link to "4 star charities." The web page also regularly publishes the list of charities with updated ratings. This list reports the star rating of charities for the previous and the current period, but does not report the actual overall rating. Furthermore, charities that were awarded a four star rating by Charity Navigator often advertise their star rating on their web page without mentioning their overall rating score.

In my empirical analysis, I focus on the period from 2007 to 2010. During this period, Charity Navigator ranked 5,400 charities annually.[5] Although the final sample is an unbalanced panel of charities, attrition rate is relatively low. In particular, I have a single year of data for 111 charities and two years of data for 345 charities. Table I shows that during this time period, most charities are rated as 3 or 4 stars (good or exceptional). Charity Navigator's ratings mainly serve as a potential information source for donors to help them make informed decisions about their charitable contributions. Charity Navigator's list includes very well-known charities such as American Red Cross, Smithsonian Institution, and American Cancer Society.[6] Although I do not have access to data on which charities are more popular or more frequently visited on Charity Navigator's web page, one possibility is that people have less information on relatively unknown and small charities and therefore, they may be more likely to view the ratings of these charities. In order to test the hypothesis that people are more likely to rely on independent third-party ratings when giving to relatively small and unknown charities, I classified the charities rated by the Charity Navigator based on their observable characteristics. There is no widely agreed upon definition of a small or large charity. In order to create subsamples of small and large charities, I consider the three main indicators of ratings in 2010 dollars: assets, fundraising expenditures, and public support measured as charitable contributions received. The median values for these variables for the full sample are 8,560,747, 397,352, and 3,845,830, respectively. At a given year, if a charity's assets, fundraising expenditures, contributions received in 2010 dollars are less than these median values, then it is classified as a small charity. Similarly, if a charity's assets, fundraising expenditures, contributions received in 2010 dollars are more than these median values, then I classify it as a large charity.[7] Table I presents a detailed distribution of charity size for each star rating. Table II presents further detailed descriptive statistics for the full sample and separately for small and large charities.

The star ratings of charities frequently change over time. In Table III, the first column shows the change in star rating for the same charity from one year to another. On average, the star rating of more than half of the charities (51.6%) changes in a calendar year relative to the previous period. This implies that there is a reasonable amount of variation in star ratings over time to test the impact of the ratings on charitable contributions. The star rating is more likely to change from one year to another among small charities (56.3%), whereas 52.2% of large charities stay at the same star in a calendar year relative to the previous year.

5. Rating data are available for three years for majority of charities rated by Charity Navigator. I include a charity in the sample if it is assigned a quality rating for 2007, 2008, 2009, or 2010.

6. The top 10 super-sized charities rated by Charity Navigator have a combined total expenses of approximately $10 billion. This list is available online at Charity Navigator's web page.

7. According to this classification, some charities are neither large nor small. I also consider an alternative classification of charities based on their assets only. Although not reported here, models which use the restricted sample from this alternative classification yield similar results compared with those reported in this paper. These results are available from the author upon request.

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Table II.
Descriptive Statistics

	Full Sample			Small Charities			Large Charities		
Star Rating	Assets	Fundraising Expenditures	Charitable Contributions	Assets	Fundraising Expenditures	Charitable Contributions	Assets	Fundraising Expenditures	Charitable Contributions
0	4,942	1,354	3,473	949	172	872	43,771	4,803	20,741
	(16,224)	(1,929)	(6,383)	(1,114)	(119)	(496)	(52,510)	(4,008)	(15,524)
1	15,576	977	4,516	2,029	177	1,177	72,693	3,962	19,651
	(108,643)	(2,475)	(12,851)	(1,987)	(107)	(653)	(305,996)	(5,879)	(32,640)
2	48,793	997	8,357	2,587	163	1,445	176,273	3,084	25,512
	(416,999)	(3,213)	(36,112)	(2,079)	(104)	(782)	(857,300)	(6,228)	(69,356)
3	115,679	1,629	16,944	2,863	165	1,656	297,903	3,900	39,765
	(887,633)	(7,315)	(69,643)	(2,112)	(97)	(809)	(1,456,417)	(11,801)	(111,410)
4	221,251	1,660	31,574	2,927	133	1,917	559,895	3,941	71,000
	(1,785,801)	(6,149)	(110,572)	(2,041)	(87)	(649)	(2,853,757)	(9,500)	(168,633)

Notes: Sample means are reported in $1,000s (2010 prices). Standard deviations in $1,000s are reported in parenthesis.

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Table III.
Interannual Changes in Star Ratings Averaged over Time

	Full Sample		Small Charities		Large Charities	
Change in Star Rating	No. of Obs.	Percent of Cumulative	No. of Obs.	Percent of Cumulative	No. of Obs.	Percent of Cumulative
−4	7	0.07	6	0.19	0	0.00
−3	64	0.63	35	1.10	8	0.25
−2	462	4.51	213	6.68	88	2.72
−1	2,241	21.90	726	22.77	704	21.79
0	4,953	48.40	1,393	43.68	1,686	52.18
1	2,114	20.66	669	20.98	653	20.21
2	361	3.53	128	4.01	87	2.69
3	31	0.30	19	0.60	5	0.15

Notes: The first column shows the change in star rating for the same charity from one year to another.

3. Methodology

I first start with estimating the effect of the overall ratings on charitable contributions received by charities. Following the earlier literature, I exploit the panel structure of the data set and estimate the following model, which incorporates time-invariant charity-level fixed effects:

$$\ln(Cont)_{ist} = \zeta' X_{ist-1} + \theta Rating_{ist-1} + \gamma_i + \mu_t + \varepsilon_{ist}. \quad (1)$$

In equation (1), $\ln(Cont)_{ist}$ represents the natural logarithm of the total amount of contributions that charity i located in state s has received at time t in 2010 dollars and ε_{ist} is the normally distributed error term. Since the amount of contributions that charities received in the current period depends on their performance and ratings in the previous period, following Okten and Weisbrod (2000) and Chhaochharia and Ghosh (2008), I use set of control variables lagged for one period. These variables are denoted by X and include fundraising expenses ($Fund$), assets ($Asset$), program service revenues (Psr), and the "price" of contributing a dollar of output to a charitable organization ($Price$).[8] Data for these control variables come from Charity Navigator. Following Okten and Weisbrod (2000), the "price" of contributing a dollar of output to a charitable organization is defined as $1/(1 - Share_{ist-1})$, where $Share_{ist-1}$ is the share of donations revenue spent by the organization on fundraising in the previous period and calculated as $Fund_{ist-1}/Cont_{ist-1}$. In this calculation, $Fund_{ist-1}$ and $Cont_{ist-1}$ denote the fundraising expenses and total amount of charitable contributions received in the previous period, respectively. Fixed charity effects are captured by γ_i, whereas μ_t represents the fixed year effects.[9] The main coefficient of interest, θ, captures the impact of the overall rating of the charity in the previous period ($Rating_{ist-1}$) on current period's total amount of charitable contributions received by the charity.

Because Charity Navigator prominently displays and heavily advertises a charity's star rating, potential donors may be more likely to pay attention to star ratings rather than the actual ratings. Furthermore, Charity Navigator's qualitative ratings change only

8. Following Okten and Weisbrod (2000) and Chhaochharia and Ghosh (2008), I use the natural logarithm of these variables that are lagged one period and expressed in 2010 dollars.

9. The location (state of residence) of all charities in the sample remains the same over time. Therefore, state-level fixed effects cannot be included to the model jointly with the charity-level fixed effects.

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at the threshold levels of overall ratings. For instance, one may expect that an increase in overall rating from 60 to 61 would have a small or statistically insignificant impact on donations since both of these ratings correspond to a four star rating and indicate that the charity is exceptional. On the other hand, an increase in actual rating from 59 to 60 would have a considerable impact on charitable donations since such an increase implies a change in the star ratings from three to four stars. This corresponds to a change from good to exceptional in qualitative ratings. In order to test this hypothesis, I use an RD design.[10] This approach exploits the sudden change in the star ratings at the threshold values of the overall ratings. Because each charity is assigned a star rating according to predetermined threshold values of overall ratings, one can compare outcomes across charities that are slightly below or above a particular threshold rating. These charities are expected to have very similar observable characteristics and actual ratings, but should differ in their star ratings. The basic RD model used through the empirical analysis is as follows:

$$\ln(Cont)_{ist} = \beta' X_{ist-1} + \alpha T_{ist-1} + f(D_{ist-1}) + \eta_s + \mu_t + \upsilon_{ist}, \tag{2}$$

where η_s denotes state fixed effects and υ_{ist} is the normally distributed error term.[11] The treatment variable is denoted by T and takes the value of unity if the charity's rating in the previous period is greater than the threshold value of the overall rating and zero otherwise. The coefficient on this variable, α, is the main coefficient of interest and measures the impact of a one star increase in the charity rating in the previous period on the amount of donation received in the current period. A smooth function of the forcing variable is denoted by $f(D_{ist-1})$, where D_{ist-1} is the difference between the overall rating and the threshold rating in the previous period.

The identifying assumption of the RD analysis in this context is that the change in the charitable contributions received at the threshold values of overall ratings should be solely due to the change in the star rating since other observable and unobservable characteristics of charities that may affect donations remain similar around the threshold values of overall ratings. This implies that except for the star rating, there should not be a significant change in other observable characteristics of charities at the threshold values of actual ratings. This is a testable assumption. In the next section, I provide the results from a formal test which shows that in equation (2), control variables vary smoothly around the threshold values of overall ratings. Hence, they have little effect on the estimates of the discontinuity and serve mainly to increase the precision of the estimates.

Modeling the smooth function of the forcing variable correctly is one of the main problems in implementing the RD design. In order to test the robustness of my results under alternative parametric model specifications, I estimate several different models that contain the first, second, or third-order polynomial of the forcing variable, which is also fully interacted with the treatment variable. For instance, in the most general model that includes a cubic polynomial of the difference between the overall rating and the

10. Imbens and Lemieux (2008), Porter (2003), and Lee and Lemieux (2009) present a detailed discussion of the RD design and related issues.

11. Inclusion of charity level fixed effects to the RD models is possible. However, because there is no reason to believe that charity-specific, time-invariant characteristics exhibit a discrete jump at the threshold values of third-party ratings, inclusion of these fixed effects should not have a significant impact on the main results.

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threshold rating that is fully interacted with the treatment variable, the smooth function of the forcing variable, $f(D_{ist-1})$, can be expressed as follows:

$$f(D_{ist-1}) = \sum_{j=1}^{k=3} \delta_j D_{ist-1}^j + \sum_{j=1}^{k=3} \lambda_j (T_{ist-1} \times D_{ist-1}^j). \tag{3}$$

In the empirical analysis, I restrict my sample to all observations in which the charity's overall rating is at most three points lower or greater than the cutoff ratings ($|D| \leq 3$). Since the RD estimates may be sensitive to the selection of this bandwidth, I also present results for alternative choices of bandwidths, that is, $|D| \leq 2$ and $|D| \leq 4$.[12]

It is also possible to estimate the discontinuity in charity ratings nonparametrically. However, this approach has two main shortcomings. First, in contrast to parametric models, it is hard to control for observable characteristics of charities in nonparametric models due to the well-known dimensionality problem. Second, one has to balance the goals of staying as local to the cutoff points at the threshold values of overall ratings as possible while ensuring to have enough data to obtain informative estimates. This requires the selection of an optimal bandwidth for the outcome variable. However, there is currently no widely agreed-upon method for selection of optimal bandwidths in the nonparametric RD context. Therefore, throughout the paper, I report results from parametric models only.[13]

4. Results

4.1. The Effect of Overall Ratings on Charitable Giving

In order to estimate the effect of a one point change in overall ratings on the amount of charitable contributions received by charities, I estimate equation (1) and report the results in Table IV. In this table, all regressions include fixed charity and year effects.[14] I also report separate results from the models that are estimated with and without a set of control variables. For charities that are rated as 4 stars, the estimation results imply that a one point increase in overall ratings is associated with a 1.3% decrease in charitable contributions received. However, this effect becomes positive and statistically insignificant once the control variables are included in the model. For charities that are rated as 3 stars, the estimation results show that an increase in overall ratings does not have a significant impact on charitable contributions received by charities. On the other hand, for charities that are rated as 2 stars. a one point increase in overall ratings is associated with a 1.1–1.5% decrease in charitable contributions received.

12. In addition to panel data models, Grant (2010) uses a simple RD design as a robustness check when estimating the effect of third-party ratings on charitable giving. In contrast to Grant (2010), I use a more flexible functional form assumption and consider the first, second, and third-order polynomial of the forcing variable in RD models. Moreover, I test whether the results are sensitive to the selection of alternative bandwidths or sample based on the set of control variables.

13. Although not reported here, I also estimate nonparametric models following Hahn et al. (2001). In these models, I use triangular kernel which has been shown to be boundary optimal by putting more weight on observations closer to the cutoff point (Cheng et al., 1997) and employ the bandwidth selection procedure suggested by Imbens and Kalyanamaran (2012). In general, nonparametric models yield a smaller impact of a one star increase in ratings on charitable contributions. Furthermore, these effects remain to be statistically insignificant. These results are available from the author upon request.

14. If there exists time-variant unobservable charity characteristics that jointly affect charitable contributions and overall ratings, the panel data analysis using charity level and year fixed effects may yield a biased estimate of the impact of the overall ratings on charitable contributions received by charities.

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Table IV.
The Effect of the Overall Ratings on Charitable Contributions

	Full Sample		Small Charities		Large Charities	
Four star						
Rating	−0.013	0.000	−0.013	−0.002	−0.008	0.002
	(0.004)***	(0.004)	(0.007)*	(0.007)	(0.006)	(0.006)
R^2	0.059	0.112	0.097	0.147	0.065	0.122
No. of obs.	3,457	3,457	1,007	1,007	1,276	1,276
Controls	No	Yes	No	Yes	No	Yes
Three star						
Rating	−0.001	0.004	−0.009	−0.005	0.001	0.007
	(0.003)	(0.003)	(0.006)	(0.005)	(0.005)	(0.005)
R^2	0.034	0.087	0.043	0.093	0.114	0.166
No. of obs.	3,705	3,701	1,052	1,051	1,311	1,311
Controls	No	Yes	No	Yes	No	Yes
Two star						
Rating	−0.015	−0.011	−0.006	−0.005	−0.009	0.002
	(0.005)***	(0.006)*	(0.008)	(0.008)	(0.011)	(0.011)
R^2	0.038	0.070	0.057	0.069	0.177	0.299
No. of obs.	2,228	2,224	816	815	537	535
Controls	No	Yes	No	Yes	No	Yes

Notes: The effect of the overall (actual) rating on charitable contributions received by charities is reported. All regressions include fixed charity and year effects. Control variables are lagged for one period and include fundraising expenses, assets, program service revenues, and price of contributing a dollar of output to a charitable organization. These variables are discussed in the text in detail. Robust standard errors clustered at the charity level are reported in parenthesis. The signs * and ** indicate statistical significance at 10% and 1% levels, respectively.

Compared with the results for the full sample, the results for small or large charities are quite similar. In general, for small and large charities, within the same star or quality ratings, overall ratings have no significant impact on charitable contributions received. This result implies that although Charity Navigator displays both the overall and star rating of charities, potential donors do not appear to be affected by the overall ratings. However, star ratings may still play an important role in giving decision. In the next section, I test this hypothesis by using an RD design, which exploits the sudden change in the star ratings at the threshold values of the overall ratings.

4.2. The Effect of Star Ratings on Charitable Giving

As mentioned before, the main identifying assumption to estimate RD type models is that the change in the charitable contributions received by charities at the threshold values of overall ratings should be solely due to the change in the star rating. This is because if other observable or unobservable characteristics of charities that may affect donations exhibit a discrete jump at the threshold values of overall ratings, then it is impossible to determine whether the change in charitable contributions received by charities is due to the change in the star rating or change in some other indicator of overall ratings. This identifying assumption implies that except for the star rating, there should not be a significant change in other observable characteristics of charities at the threshold values of actual ratings. Therefore, I first test the possibility that other changes in observable characteristics of charities may occur at the threshold values of the overall ratings. In the RD context, this is equivalent to testing the smoothness of all control variables around the threshold values of the overall ratings. Hence, I estimate equation (2) separately for

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Table V.
Test of the Smoothness of Control Variables around the Threshold Ratings

	Asset($t-1$)	Fund($t-1$)	Price($t-1$)	Psr($t-1$)
Full sample	0.291	−0.081	−0.006	0.136
	(0.195)	(0.148)	(0.017)	(0.680)
	[5,396]	[5,396]	[5,387]	[5,396]
Small charities	0.052	−0.233	0.031	−1.354
	(0.181)	(0.163)	(0.024)	(1.128)
	[1,674]	[1,674]	[1,670]	[1,674]
Large charities	−0.065	−0.186	0.026	−0.165
	(0.261)	(0.190)	(0.020)	(1.165)
	[1,705]	[1,705]	[1,704]	[1,705]

Notes: Charities with an overall rating of at most three points lower or greater than the cutoff ratings are used in all regressions. All regressions include a flexible third-order polynomial of D that is fully interacted with the treatment variable, where D is the difference between the actual and the cutoff rating. Robust standard errors clustered at the charity level are reported in parenthesis. Number of observations is reported in brackets.

all control variables using a cubic polynomial of forcing variable that is fully interacted with the treatment.[15] In Table V, I report the results separately for the full sample and for small and large charities. The coefficient of the treatment variable is insignificant in all regressions and hence, there is no evidence of a significant change in control variables at the threshold values of the overall ratings. For the full sample, I also graph the cubic prediction of each covariate fitted on actual observations on either side of the cutoff rating in Figure 1. These predictions appear to fit the actual data well and exhibit either no or an insignificant small jump at the cutoff ratings. Although, one cannot directly test whether the unobservable characteristics of charities vary smoothly across the discontinuity, the finding that observable characteristics are smoothly distributed around the cutoff ratings reduces the concerns about omitted variables bias and suggests that parametric models estimated with or without controls should yield similar results.

Another concern for the application of RD design in the context of the third-party ratings is the possibility that the forcing variable (a function of overall rating) is being manipulated. In particular, since the ratings are public knowledge, it is possible that nonprofits systematically manipulate their financial information to achieve a higher Charity Navigator rating. If this is the case, then RD estimates of the effect of ratings on charitable contributions received may be biased. However, if charities manipulate their financial information to achieve a higher rating, then one expects to observe a discontinuity in the distribution of charities at the threshold levels of ratings. I investigate this possibility in Figures 2–4. Figure 2 shows that for the full sample, the number of charities that are just below a cutoff rating is very similar to the number of charities that are just above a cutoff rating and there is no evidence of a discrete jump in the number of charities around the threshold levels of the third-party ratings. Furthermore, I also test the possibility of nonrandom sorting of charities around the threshold ratings due to the manipulation of the forcing variable using a formal test suggested by McCrary (2008). This test is based on an estimator for the discontinuity at the cutoff ratings in the density function of the forcing variable. The test is implemented as a Wald test

15. In these models, I restrict the sample to all observations in which a charity's overall rating is at most three points lower or greater than the threshold values of the overall ratings. Estimating models separately for all control variables using a linear or quadratic polynomial of forcing variable yields similar results. These results are available from the authors upon request.

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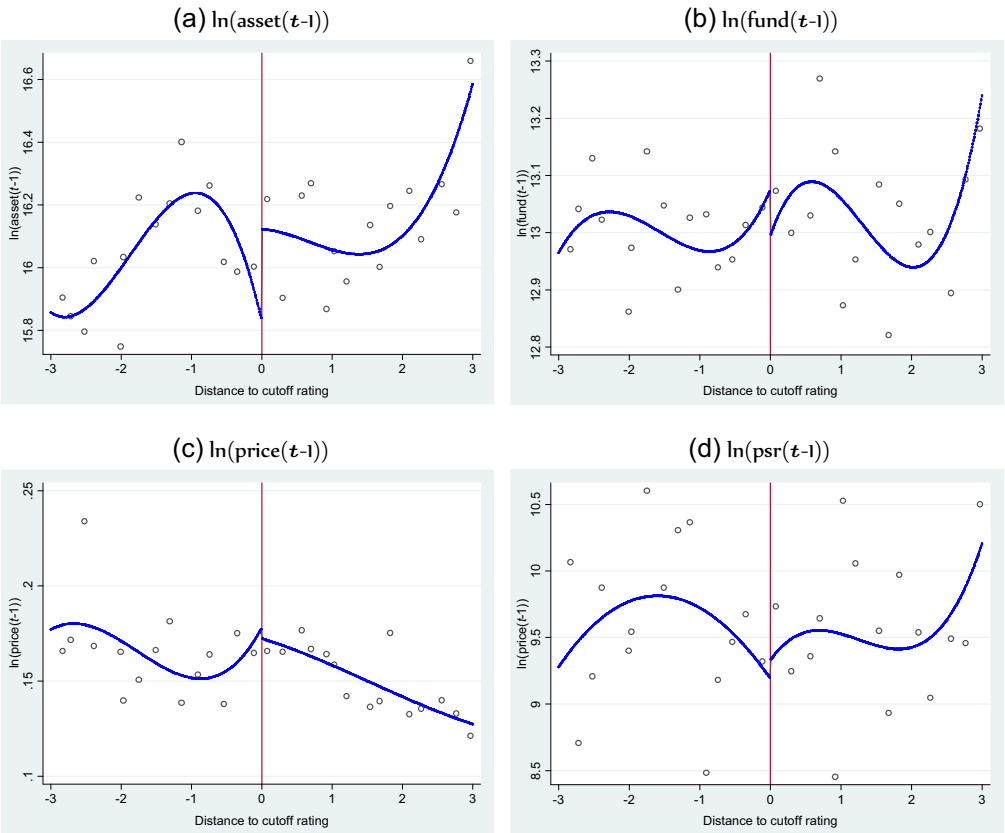

FIGURE 1. TRENDS IN SELECTED OBSERVABLE COVARIATES AROUND THE CUT-OFF RATING
Note: Each dot represents the mean of all observations (charities) that have a rating within a 0.2 point interval. The solid lines are a third-order polynomial fitted on actual observations on either side of the cutoff rating.

of the null hypothesis that the discontinuity is zero. McCrary's density test yields a discontinuity estimate of 0.076 with a standard error of 0.082. Therefore, I fail to reject the null hypothesis that there exists a nonrandom sorting of charities across cutoff ratings and conclude that the RD design should yield creditable estimates of the impact of a one star increase in ratings on charitable contributions received by charitable organizations. Similarly, Figures 3 and 4 show that for small and large charities, there is no evidence of a discrete jump in the number of charities around the threshold levels of the third-party ratings. This finding implies that the RD design should also yield creditable estimates of the impact of a one star increase in ratings on charitable contributions received by small or large charities.

4.2.1. Full Sample Results

Table VI reports the impact of a one star increase in the third-party ratings on charitable contributions received by charities. Results from the main model, which uses data from observations in which a charity's overall rating is at most three points lower or greater than the cutoff ratings ($|D| \leq 3$) are reported in the second column. The most flexible

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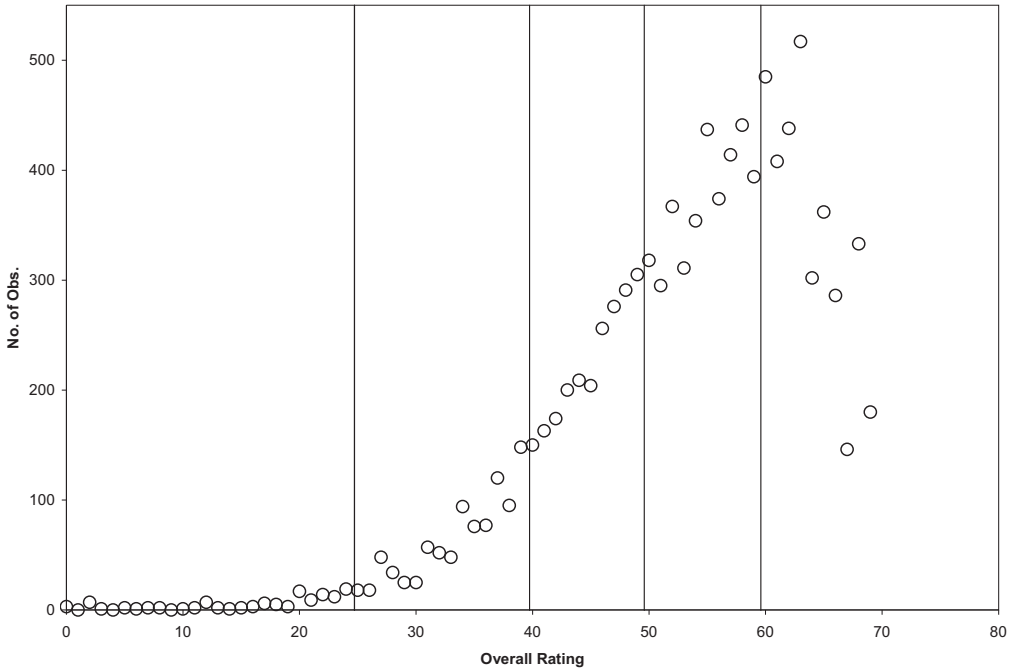

FIGURE 2. DISTRIBUTION OF CHARITIES AROUND CUTOFF RATINGS: FULL SAMPLE
Note: Total number of charities for each overall rating point interval are plotted. The solid lines show the cutoff ratings at overall ratings of 25, 40, 50, and 60.

TABLE VI.
THE EFFECT OF STAR RATINGS ON CHARITABLE CONTRIBUTIONS FOR THE FULL SAMPLE

	The Difference between the Actual and the Cutoff Rating		
	$\lvert D\rvert \leq 2$	$\lvert D\rvert \leq 3$	$\lvert D\rvert \leq 4$
Linear	0.015	−0.006	0.014
	(0.049)	(0.039)	(0.034)
Quadratic	0.095	0.032	0.016
	(0.067)	(0.058)	(0.051)
Cubic	0.062	0.126	0.051
	(0.088)	(0.074)*	(0.065)
No. of obs.	3560	5387	7152

Notes: The regressions include a set of control variables as discussed in the text and a flexible first, second, or third-order polynomial of D that is fully interacted with the treatment variable, where D is the difference between the actual and the cutoff rating. Robust standard errors clustered at the charity level are reported in parenthesis. The sign * indicates statistical significance at 10% .

model estimated using a flexible third-order polynomial of the forcing variable that is interacted with the treatment variable suggest that a one star increase in the third-party ratings is associated with a 12.6% increase in charitable contributions. This effect is marginally significant at 10% significance level. Panel A of Figure 5 illustrates this result and shows a discrete jump in the contribution amount at the cutoff rating. However, this effect is not robust to the selection of alternative bandwidths. When the sample

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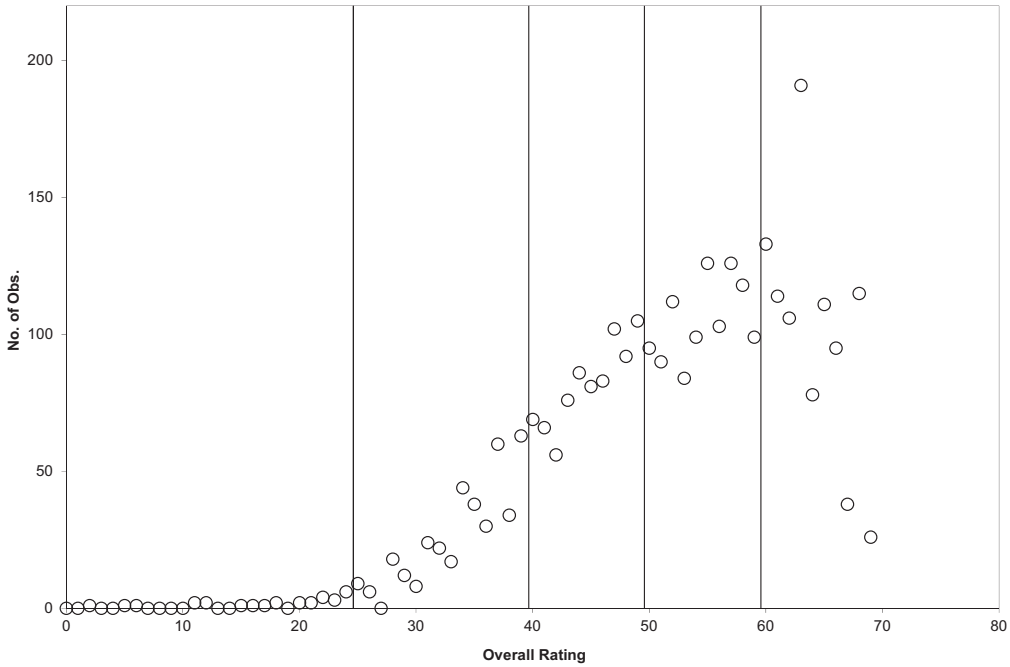

FIGURE 3. DISTRIBUTION OF CHARITIES AROUND CUTOFF RATINGS: SMALL CHARITIES
Note: Total number of charities for each overall rating point interval are plotted. The solid lines show the cutoff ratings at overall ratings of 25, 40, 50, and 60.

is restricted to observations in which a charity's overall rating is at most two or four points lower or greater than the cutoff ratings ($|D| \leq 2$ or $|D| \leq 4$), the estimated effect of ratings on charitable contributions decreases considerably and becomes statistically insignificant. Furthermore, the estimated impact of ratings on charitable contributions is not robust to selection of alternative model specifications. Results reported in the first and the second rows of Table VI show that models that contain the first or second-order polynomial of the forcing variable yield a smaller and statistically insignificant effect of ratings on charitable contributions.

It is also plausible to expect that the effect of an increase in star ratings would be different for charities with different star ratings. Suppose that a charity is rated as 1 star in the first period and as 2 stars in the second period. Although the star rating of this charity has increased by one star, the effect of this increase on charitable contributions would be relatively small since a two star rating still indicates that the charity is not operating efficiently. However, an increase in the star rating from two to three stars or two to four stars would significantly increase the contributions received by the charity since a three or four star rating indicates that the charity performs as well as or better than most charities in its cause. In order to test whether the effect of a one star increase in the star rating is different for charities with different star ratings, I estimate equation (2) separately for charities that are around two, three, or four star rating cutoffs and report the results from the most flexible models that contain a cubic polynomial of the forcing variable in Table VII. The results from the main model for which $|D| \leq 3$ show that a one star increase from 3 to 4 stars is associated with a 14.5% increase in the amount of

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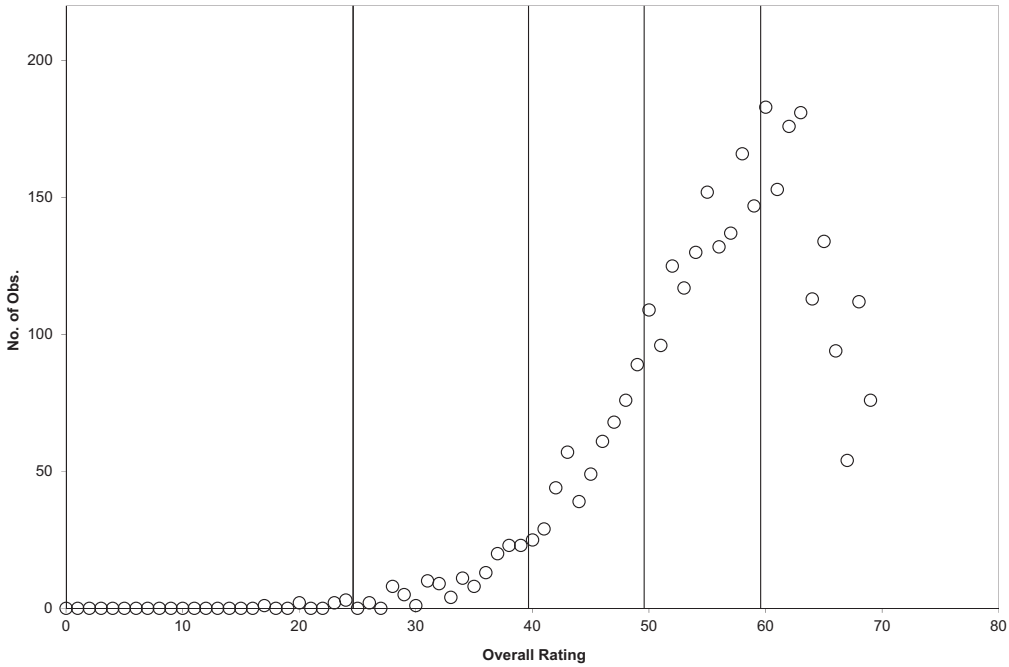

FIGURE 4. DISTRIBUTION OF CHARITIES AROUND CUTOFF RATINGS: LARGE CHARITIES
Note: Total number of charities for each overall rating point interval are plotted. The solid lines show the cutoff ratings at overall ratings of 25, 40, 50, and 60.

TABLE VII.
THE EFFECT OF STAR RATINGS ON CHARITABLE CONTRIBUTIONS AROUND ALTERNATIVE STAR RATINGS FOR THE FULL SAMPLE

	The Difference between the Actual and the Cutoff Rating		
	$\lvert D\rvert \leq 2$	$\lvert D\rvert \leq 3$	$\lvert D\rvert \leq 4$
4 star	0.112	0.145	0.051
	(0.112)	(0.094)	(0.079)
	[1,729]	[2,583]	[3,474]
3 star	−0.053	0.021	0.037
	(0.141)	(0.113)	(0.103)
	[1,214]	[1,856]	[2,420]
2 star	0.002	0.207	0.116
	(0.256)	(0.194)	(0.165)
	[552]	[850]	[1,122]

Notes: All regressions include a set of control variables as discussed in the text and a flexible third-order polynomial of D that is fully interacted with the treatment variable, where D is the difference between the actual and the cutoff rating. Robust standard errors clustered at the charity level are reported in parenthesis. The number of observations is reported in brackets.

charitable contributions. However, this effect is neither statistically significant nor robust to the selection of alternative bandwidths such as $|D| \leq 2$ or $|D| \leq 4$. Because the sample is restricted to those charities that have an initial rating around the 4 star cutoff, the threshold value of the overall ratings is 60. Panel B of Figure 5 plots the predicted value

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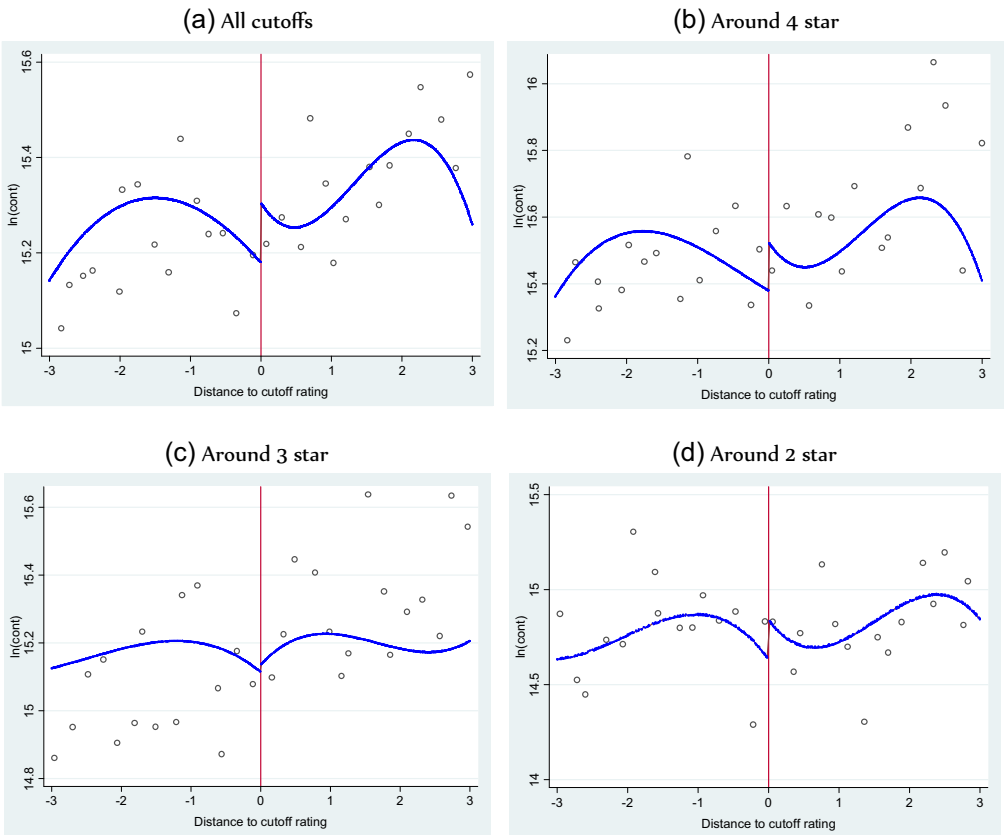

FIGURE 5. TRENDS IN CONTRIBUTIONS RECEIVED AROUND THE CUTOFF RATING: FULL SAMPLE
Note: Each dot represents the mean of all observations (charities) that have a rating within a 0.2 point interval. The solid lines are a third-order polynomial fitted on actual observations on either side of the cutoff rating.

of the outcome variable and shows that there is a small and statistically insignificant jump in charitable donations at this threshold rating. Estimating equation (2) separately for charities that are around the 2 or 3 star cutoffs also yields similar results. Under alternative models that use different bandwidths, the effect of the third-party ratings on the amount of charitable contributions remains statistically insignificant. Panels C and D in Figure 5 also show that charitable contributions received by the charities that are slightly below the threshold overall ratings of 40 (2 stars) and 50 (3 stars) are not statistically different from those charities that are slightly above these threshold ratings.

Using a sample of charities rated by Charity Navigator, Grant (2010) uses an RD design as a robustness check to estimate the effect of the star ratings on charitable contributions received by charities that are assigned different star ratings. Compared with my models for charities with different star ratings, her RD model uses a larger bandwidth ($|D| \leq 5$) and less flexible functional form that contains the first-order polynomial of the difference between the overall rating and the threshold rating. She finds that an increase in star ratings from 0 to 1 star has no impact on charitable contributions received by the

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Table VIII.
The Effect of Star Ratings on the Amount of Charitable Contributions for Small Charities

	The Difference between the Actual and the Cutoff Rating		
	$\lvert D\rvert \leq 2$	$\lvert D\rvert \leq 3$	$\lvert D\rvert \leq 4$
Linear	0.164	0.121	0.073
	(0.058)***	(0.048)**	(0.041)*
Quadratic	0.168	0.192	0.187
	(0.083)**	(0.069)***	(0.059)***
Cubic	0.184	0.195	0.235
	(0.110)*	(0.088)**	(0.077)***
No. of obs.	1,101	1,670	2,253

Notes: The regressions include a set of control variables as discussed in the text and a flexible first, second, or third-order polynomial of D that is fully interacted with the treatment variable, where D is the difference between the actual and the cutoff rating. Robust standard errors clustered at the charity level are reported in parenthesis. The signs *, **, and *** indicate statistical significance at 10% , 5% , and 1% , respectively.

charity. On the other hand, her findings imply that charities that are slightly below the one, two, or three star rating thresholds may observe 3–5% increase in charitable contributions if they manage to increase their ratings by one star. In terms of the magnitude of the impact, the estimates reported in the last column of Table VII are comparable to that of Grant (2010). However, my estimates, which are from an RD model that uses a more conservative bandwidth and more flexible functional form, are insignificant.

4.2.2. Results for Small and Large Charities

In a recent paper, Luca (2011) shows that a one star increase in Yelp.com rating leads to a 5–9% increase in revenues of restaurants. However, this effect is primarily driven by independent restaurants and Yelp.com ratings do not affect restaurants with chain affiliation. This finding implies that online consumer reviews substitute for more traditional forms of reputation if the product is relatively unknown and people cannot easily observe its quality. Compared with the rest of the sample, small charities have relatively limited resources to signal their quality through fundraising or advertising. If Luca's finding also applies to nonprofit sector, it is possible that potential donors put more value on the third-party ratings when giving to small charities. In Table VIII, I report the effect of a one star increase in ratings on charitable contributions received by small charities. The results from the most flexible parametric model that contains a cubic polynomial of the forcing variable indicate that for this group of charities, a one star increase in ratings is associated with a 19.5% increase in the amount of contributions received. This effect is highly significant and robust to the selection of alternative bandwidths. Furthermore, the estimated impact of the third-party ratings on charitable contributions received is robust under alternative model specifications. Models estimated using a quadratic polynomial of the difference between the overall rating and the threshold rating yield similar results compared with the models estimated using a cubic polynomial of the forcing variable. Although models estimated using a linear polynomial of the forcing variable yield a relatively smaller effect of the ratings on charitable donations, this effect remains to be statistically significant, positive and compared with the full sample, considerably larger. Panel A of Figure 6 also illustrates this result and shows that charitable contributions exhibit a discrete jump at the cutoff ratings for small charities.

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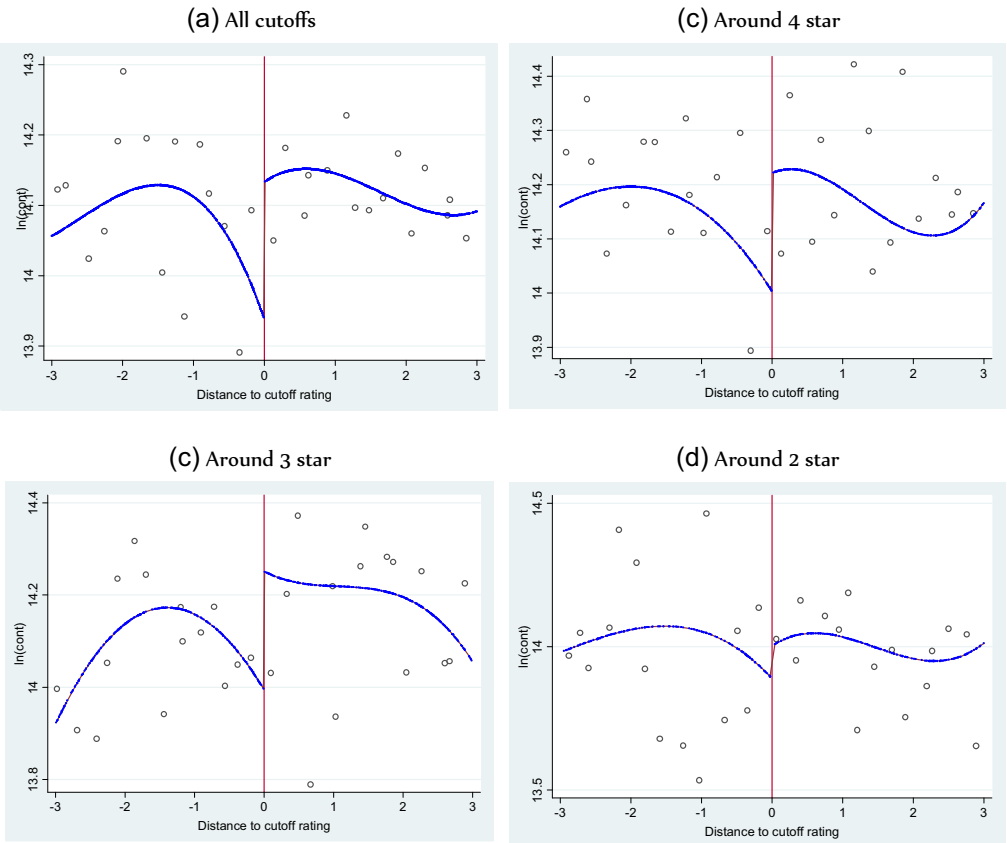

FIGURE 6. TRENDS IN CONTRIBUTIONS RECEIVED AROUND THE CUTOFF RATING: SMALL CHARITIES
Note: Each dot represents the mean of all observations (charities) that have a rating within a 0.2 point interval. The solid lines are a third-order polynomial fitted on actual observations on either side of the cutoff rating.

The effect of an increase in star ratings is also different for small charities with different initial star ratings. Table IX shows that for small charities that are slightly below the 2 star threshold rating, a one star increase has no significant impact on the amount of contributions that the charity would receive. This result is not surprising because an increase from 1 star to 2 stars implies that the charity was able to increase its quality rating from "exceptionally poor" to "poor," but it is still rated well below many charities that are rated as "good" (3 stars) or "exceptional" (4 stars). Therefore, an increase from 1 to 2 stars in ratings would still signal that the charity is below an acceptable quality level. However, for small charities that are slightly below the three or four star rating threshold, a one star increase in the rating is associated with up to a 28.2% increase in the amount charitable donations under alternative models. This result implies that only relatively higher ratings (3 or 4 stars) has a positive signaling value for small charities and lead to increased charitable donations. Figure 6 also documents this result. In Panels B and C, the amount of charitable contributions exhibit a discrete jump at the threshold ratings of 4 and 3 stars, respectively. However, Panel D shows that the

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Table IX.
The Effect of Star Ratings on the Amount of Charitable Contributions around Alternative Star Ratings for Small Charities

	The Difference between the Actual and the Cutoff Rating		
	$\lvert D\rvert \leq 2$	$\lvert D\rvert \leq 3$	$\lvert D\rvert \leq 4$
4 star	0.198	0.218	0.251
	(0.142)	(0.119)*	(0.104)**
	[466]	[697]	[991]
3 star	0.246	0.257	0.282
	(0.178)	(0.139)*	(0.118)**
	[383]	[596]	[764]
2 star	0.141	0.114	0.239
	(0.318)	(0.208)	(0.186)
	[230]	[347]	[452]

Notes: All regressions include a set of control variables as discussed in the text and a flexible third-order polynomial of D that is fully interacted with the treatment variable, where D is the difference between the actual and the cutoff rating. Robust standard errors clustered at the charity level are reported in parenthesis. The number of observations is reported in brackets. The signs * and ** indicate statistical significance at 10% and 5% , respectively.

Table X.
The Effect of Star Ratings on Charitable Contributions for Large Charities

	The Difference between the Actual and the Cutoff Rating		
	$\lvert D\rvert \leq 2$	$\lvert D\rvert \leq 3$	$\lvert D\rvert \leq 4$
Linear	−0.066	−0.083	−0.079
	(0.073)	(0.060)	(0.051)
Quadratic	−0.108	−0.101	−0.080
	(0.099)	(0.086)	(0.076)
Cubic	−0.065	−0.096	−0.121
	(0.130)	(0.106)	(0.093)
No. of obs.	1,128	1,704	2,263

Notes: The regressions include a set of control variables as discussed in the text and a flexible first, second or third-order polynomial of D that is fully interacted with the treatment variable, where D is the difference between the actual and the cutoff rating. Robust standard errors clustered at the charity level are reported in parenthesis.

amount of charitable contributions received by charities that are slightly above the two star rating is not statistically different from those that are slightly below the two star rating.

In Table X, I report the impact of star ratings on charitable contributions received by large charities. Regardless of the bandwidth or model selection, the effect of a one star increase in star ratings on charitable contributions received by large charities is negative. In particular, the coefficient estimates show that a one star increase in ratings is associated with a 6.5–12.1% decrease in charitable contributions received by large charities. However, these estimates are not statistically significant at conventional significance levels. Panel A of Figure 7 also illustrates this result. Furthermore, Table XI shows that for large charities, the effect of star ratings on charitable contributions remains to insignificant once separate models are estimated for large charities with different initial star ratings. Panels B, C, and D of Figure 7 also document that for large charities, the discrete change in charitable contributions at the threshold star ratings is small and statistically insignificant.

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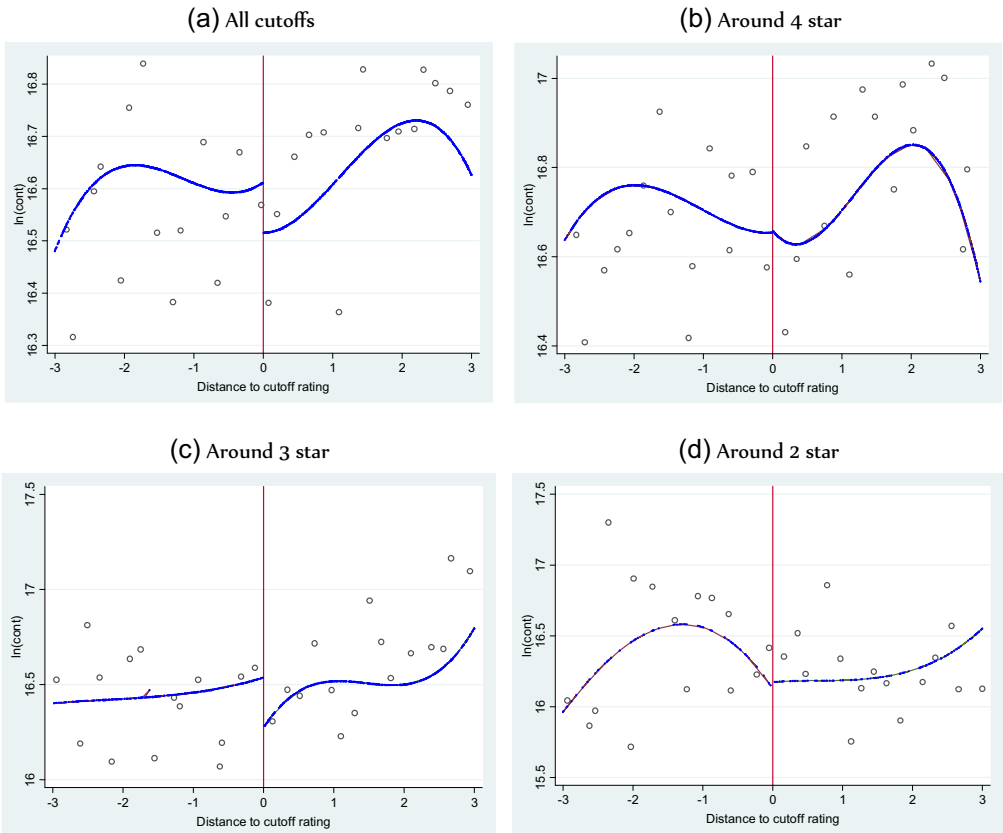

FIGURE 7. TRENDS IN CONTRIBUTIONS RECEIVED AROUND THE CUTOFF RATING: LARGE CHARITIES

Note: Each dot represents the mean of all observations (charities) that have a rating within a 0.2 point interval. The solid lines are a third-order polynomial fitted on actual observations on either side of the cutoff rating.

Table XI.
The Effect of Star Ratings on Charitable Contributions around Alternative Star Ratings for Large Charities

	The Difference between the Actual and the Cutoff Rating		
	$\lvert D\rvert \leq 2$	$\lvert D\rvert \leq 3$	$\lvert D\rvert \leq 4$
4 star	0.086	0.005	−0.090
	(0.162)	(0.129)	(0.110)
	[649]	[964]	[1,275]
3 star	−0.401	−0.265	−0.217
	(0.248)	(0.187)	(0.164)
	[373]	[566]	[740]
2 star	0.253	0.045	0.159
	(0.565)	(0.322)	(0.267)
	[99]	[164]	[233]

Notes: All regressions include a set of control variables as discussed in the text and a flexible third-order polynomial of D that is fully interacted with the treatment variable, where D is the difference between the actual and the cutoff rating. Robust standard errors clustered at the charity level are reported in parenthesis. The number of observations is reported in brackets.

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4.2.3. Robustness Checks

In all regressions, I use a set of control variables lagged for one period. These variables include fundraising expenses, assets, program service revenues, and the price of contributing a dollar of output to a charitable organization. However, it is also possible that while current period's donations may be affected by several efficiency measures from the previous period, compared with previous period's fundraising expenses, contemporary fundraising expenses of charities may be more important in giving decisions of potential donors. In order to test this possibility, I estimate models in which in addition to other control variables, I control for current period's fundraising expenses instead of fundraising expenses lagged for one period.[16] The results reported for the full sample in the first specification of Table XII shows that using current period's fundraising expenses as a control variable instead of fundraising expenses lagged for one period does not affect the estimated impact of a one star increase in ratings on charitable contributions. Furthermore, specifications 4 and 7 of Table XII show that compared with the original results, results for the small and large charities remain virtually the same even after current period's fundraising expenses are controlled for in the RD models.

In specifications 2, 5, and 8 of Table XII, I investigate whether the main results are driven by some outlier observations. I first drop the charities that are at the top or bottom 10 percentile of the sample based on their fundraising expenses, assets, and charitable contributions received and reestimate the models with a restricted sample of charities.[17] The results reported in the second specification of Table XII show that the effect of the star ratings on charitable contributions remains insignificant once very small and very large charities are excluded from the sample. Next, I consider the sample of small charities and exclude those charities from this sample whose fundraising expenses, assets, and charitable contributions received are at the bottom 10 percentile.[18] The results reported in the fifth specification of Table XII document that for small charities, a one star increase ratings is associated with a 20.4% in charitable contributions received. This estimate is comparable to the main results for the small charities that are reported in Table VIII. Furthermore, the results for small charities that are around the threshold star ratings are robust to the exclusion of very small charities from the sample. In particular, the results imply that for a small charity, an increase from three to four star rating is associated with a 23.3% increase in charitable contributions. Finally, I test whether the results are robust to the exclusion of those charities that are at the top 10 percentile of the sample of large charities based on their fundraising expenses, assets, and charitable contributions received. The results reported in specification 8 of Table XII document that the effect of the star ratings on charitable contributions received by large charities remains insignificant even after very large charities are excluded from the sample.

16. In Table V, I test whether observable characteristic of the charities for the previous period exhibit a discrete jump at the threshold values of ratings. I run a similar test for current period's fundraising expenses and find that current period's fundraising expenses do not exhibit a discrete jump at the threshold values of overall ratings as well (estimated coefficient is −0.039 with a standard error of 0.149). Therefore, in the context of an RD design, current period's fundraising expenses do not have an endogenous impact when estimating the effect of star ratings on charitable contributions received by charities.

17. For the full sample, the cutoff amounts for the bottom 10 percentile of charitable contributions, fundraising expenses, and assets are $993,785, $80,281, and $1,058,670, respectively. On the other hand, the cutoff amounts for the top 10 percentile of charitable contributions, fundraising expenses, and assets are $30,271,490, $2,451,804, and $123,470,500, respectively.

18. For the sample of small charities, the cutoff amounts for the bottom 10 percentile of charitable contributions, fundraising expenses, and assets are 662,104, 35,390, and 494,492, respectively.

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Table XII.
The Effect of Star Ratings on Charitable Contributions: Robustness Checks

	All Cutoffs	4 Star	3 Star	2 Star
Full sample				
1. Use current fundraising expenses as a control variable	0.027	−0.002	0.046	0.118
	(0.066)	(0.086)	(0.108)	(0.151)
	[7,153]	[3,474]	[2,421]	[1,122]
2. Exclude charities that are at the bottom or top 10 percentile	0.073	0.053	0.003	0.150
	(0.058)	(0.079)	(0.094)	(0.144)
	[4,500]	[2,194]	[1,561]	[692]
3. Use level charitable contributions in $1,000 as a dependent variable	−8,076.54	−18,005.34	1,244.44	687.85
	(6,136.72)	(13,020.66)	(3,652.87)	(2,377.05)
	[7,152]	[3,474]	[2,420]	[1,122]
Small charities				
4. Use current fundraising expenses as a control variable	0.217	0.223	0.269	0.239
	(0.078)***	(0.112)**	(0.121)**	(0.183)
	[2,254]	[991]	[765]	[452]
5. Exclude charities that are at the bottom 10 percentile	0.204	0.234	0.144	0.361
	(0.069)***	(0.117)**	(0.111)	(0.197)*
	[1,073]	[515]	[359]	[191]
6. Use level charitable contributions in $1,000 as a dependent variable	353.31	331.52	459.29	288.60
	(109.71)***	(155.07)**	(177.57)***	(263.88)
	[2,253]	[991]	[764]	[452]
Large charities				
7. Use current fundraising expenses as a control variable	−0.101	−0.166	−0.064	0.337
	(0.092)	(0.108)	(0.164)	(0.300)
	[2,263]	[1,275]	[740]	[233]
8. Exclude charities that are at the top 10 percentile	0.059	0.039	0.085	−0.076
	(0.088)	(0.118)	(0.149)	(0.340)
	[1241]	[661]	[424]	[149]
9. Use level charitable contributions in $1,000 as a dependent variable	−21,679.23	−32,428.78	−14,063.15	16,874.42
	(17,396.99)	(27,377.6)	(13,815.48)	(12,226.23)
	[2,263]	[1,275]	[740]	[233]

Notes: In all regressions, the sample is restricted to observations in which the charity's actual rating is at most four points lower or greater than the cutoff ratings. The regressions include a set of control variables as discussed in the text and a third-order polynomial of D that is fully interacted with the treatment variable, where D is the difference between the actual and the cutoff rating. Robust standard errors clustered at the charity level are reported in parenthesis. Number of observations is reported in brackets. The signs *, **, and *** indicate statistical significance at 10% , 5% , and 1% , respectively.

In my models, I use the natural logarithm of the total amount of contributions received by charities as a dependent variable. The results reported in Table XII show that using the actual dollar amount of contributions rather the log transformed variable does not affect the main findings. In particular, the results for the full sample and large charities remain insignificant, whereas the estimated impact of star ratings on charitable contributions received by small charities is highly significant. For small charities, the sixth specification in Table XII shows that a one star increase in ratings is associated with a slightly more than 353,000 increase in charitable contributions received. A significant and positive impact of a one star increase in ratings is also observed for small charities that are ranked as 2 or 3 stars.

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5. Conclusion

This paper investigates the effect of the third-party charity ratings on charitable behavior using a large sample of charities rated by Charity Navigator, the largest independent charity evaluator in the United States. First, I estimate several panel data fixed effects models that control for unobserved time-invariant charity characteristics and find that an increase in overall ratings does not have a significant impact on charitable contributions received by charities that have the same star rating. This result implies that potential donors do not appear to be affected by the overall ratings. However, star ratings that change at the threshold values of actual ratings may still play an important role in giving decision. In order to test this possibility, I use an RD design and investigate whether charitable contributions received by charities exhibit a discrete jump at the threshold values of actual ratings.

For the full sample, I document that a one star increase in Charity Navigator ratings has no significant effect on the amount of charitable contributions received by charities. A similar result also observed for large charities. However, the third-party ratings have a considerable impact on the amount of charitable donations received by relatively small charities whose quality are less likely to be identified by charitable donors. For this group, a one star increase in ratings is associated with a 19.5% increase in the amount of charitable contributions received. The effect of an increase in the star rating is considerably higher for those charities that have a relatively higher rating. For small charities that are slightly below the two star rating, a one star increase has no significant impact on the amount of contributions that the charity would receive. However, for small charities, a one star increase from 2 to 3 or 3 to 4 stars is associated with up to a 28.2% increase in the amount charitable donations received. These results are also robust under alternative model specifications.

The findings of this paper contribute to the existing literature, which documents the importance of information in the charity market.[19] The main finding that donors respond to signals regarding the quality of relatively small charities has also important policy implications. Several recent papers have shown the positive impact of mandatory quality disclosure laws. For instance, Jin and Leslie (2003) show that a restaurant whose hygiene report card grade moves from a B to an A experiences a 5% increase in revenue relative to other grades. Bollinger et al. (2011) find that mandatory calorie posting laws cause consumers to consume 6% fewer calories at a leading chain restaurant. The third-party ratings are similar to mandatory quality disclosure laws in a sense that they are both designed to provide information to consumers, so that they can make informed decisions and hold firms accountable for their actions. Therefore, the findings of this paper imply that a policy that will enforce charities to publicly announce their quality ratings may help charitable donors to make informed decisions about their contributions and may also work as an advertising mechanism for relatively small charities that have above average quality as rated by independent charity evaluators. In particular, such a policy would potentially help small charities that are well-managed and receive good ratings, but do not have resources to spend money on fundraising or advertising. However, further research is needed to understand the relationship between quality disclosure laws and independent ratings and to investigate charities' response to the third-party ratings.

19. There is a considerable amount of literature which argue that fundraising provides information regarding the quality of charities. See, for example, Rose-Ackerman (1982), Andreoni (2006), Yörük (2009, 2009, 2012a,b), and Mungan and Yör ük (2012).

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